

Mechanical Important Topic

Unit 1

Thermodynamics law's

Thermal and nuclear power plant

Boiler- water tube and fire tube

Classification of boiler - Cochran, locomotive, Wilcox,

Heat engine

Unit 3

Refrigeration cycle -

* VCR (vapor compression refrigeration)

* VAR(vapor absorption refrigeration) working and components

Air conditioning system

Split and window AC

Unit 2

IC engine , components working

Four stroke and 2 stroke engine

SI and CI engine (difference)

Pump- centrifugal and reciprocating pump(single acting and double acting)

Unit 4

Types of gear , belt , belt drives and gear trains

Length of belt. And slip of belt and tension ratio derivation

Unit 5

* Casting - definition, tools , steps = molding pouring cooling

* metal forming - forging, rolling , extrusion and drawing and tools used

* metal joining- welding (gas and arc welding), brazing , soldering,

Unit 6

Properties of materials

Classification of materials

Heating treatment- annealing, normalizing, hardening, tempering, case hardening